

KshetraAI

Explainable Field-Force Intelligence

For agricultural sales operations

Static visit lists → adaptive, explainable, measurable execution

WHO

Prioritized daily visits for the field representative.

WHAT

Next best action, alert context, and visible reasoning.

WHY IT MATTERS

The prototype turns generated intelligence into a usable field workflow while keeping humans in control.

Static plans cannot keep up with field reality

Inventory, demand, crop context, and visit outcomes change faster than manual planning cycles.

FIELD REALITY

Inventory movement, demand, crop context, and retailer urgency shift quickly.

MANUAL PLANNING

Fixed visit schedules do not explain why one visit matters more than another.

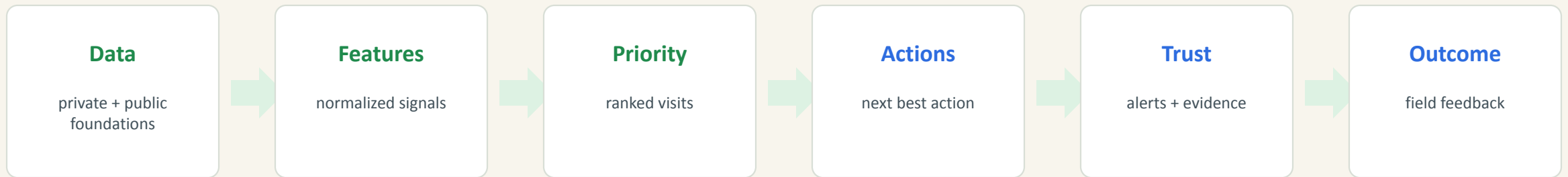
MISSING LOOP

Outcomes are often not captured back into the operating system.

The core question: who should the rep visit, what should they do, and why?

A deterministic workflow from signals to outcomes

KshetraAI connects data, feature signals, prioritization, actions, alerts, explanations, and outcome capture.



DESIGN PRINCIPLE

Deterministic and explainable by default. Humans own the architecture and business logic.

Grounded data foundation

Private raw files stay local; processed views and sanitized API outputs drive the demo.

INTERNAL DATA USED

- Representative / territory data
- Retailers and growers
- Retailer point-of-sale
- Weekly inventory
- Visit logs
- Digital funnel and WhatsApp engagement

PUBLIC + CONTROLLED SIGNALS

- Weather is usable public signal data
- Crop context is curated
- NDVI is metadata/reference level
- Pest data is reference-level, not live outbreak detection

Privacy boundary: raw private data stays local; demo uses derived processed outputs and sanitized API-level samples.

Modular intelligence layers

Each layer has a focused job, test coverage, and clear boundaries.

18

registered features

10k

feature rows generated

8,827

alert rows generated

6

demo explanations

MODULES

Feature generation → priority ranking → next best action → alerts → explainability → outcome foundation

How the system solves the workflow

Each component turns a raw operational question into a traceable decision step.

Data Foundation

- Logic: validate + join data
- Unlocks: trusted views

Feature Builder

- Logic: normalize signals
- Unlocks: explainable scores

Priority Engine

- Logic: weighted ranking
- Unlocks: focused visits

Next Best Action

- Logic: deterministic rules
- Unlocks: field guidance

Alert Engine

- Logic: baseline checks
- Unlocks: early risk signals

Explainability

- Logic: evidence mapping
- Unlocks: trust

Outcome Layer

- Logic: capture feedback
- Unlocks: learning foundation

API + UI

- Logic: expose workflow
- Unlocks: usable demo path

Ground truth: each box maps to implemented modules, API/UI surfaces, or verified demo artifacts.

Daily plan prioritization in the demo

For the fixed scenario, the top ranked entity is RTL_01300 because inventory need is the strongest signal.

FIXED SCENARIO

REP_0164
TER_0164
2026-05-17
Top entity: RTL_01300

#1

rank in daily plan

37.1

priority score

Low

absolute priority level

Main reason: Highest signal is inventory need (92.5). Current scores are low by absolute threshold, but relative ranking still identifies the first visit.

Actionable recommendation with visible evidence

The selected entity gets a next best action, active alert context, and explanation evidence.

NEXT BEST ACTION

- Possible fast-moving stock pressure
- Relevant Seasonal SKU
- Medium confidence
- Actions: review availability, check reorder timing, plan follow-up

ACTIVE ALERT

- Possible stock-out risk
- Severity score: 78.5
- Severity: High
- Confidence: High

EXPLANATION

- 6 explanation rows
- Types: priority, recommendation, anomaly
- Evidence and confidence remain visible

RTL_01300

Closing the field loop

The outcome layer captures what happened after the recommendation and keeps recalibration human-reviewed.

CAPTURED FIELDS

- Visit completed
- Recommendation followed
- Sale made / order placed
- Order value
- Alert validation
- Rep feedback

CURRENT TRUTH

- API normalizes one submitted outcome
- Sample response returns success
- No persistent historical outcome dataset artifact yet
- Recalibration is human-review only

Outcome ID: OUTCOME_RTL_01300_POSSIBLE_FAST_MOVING_STOCK_PRESSURE

Thin API, focused frontend workflow

FastAPI exposes generated intelligence; React turns it into a field workflow.

FASTAPI BACKEND

- GET /daily-plan
- GET /recommendations/{entity_id}
- GET /alerts
- GET /explanations/{entity_id}
- POST /outcomes
- Swagger via FastAPI defaults



REACT FRONTEND

- Overview selection
- Daily plan
- Recommendation + explanation
- Alert review
- Outcome submission
- UI does not compute intelligence

API reads processed outputs; frontend displays backend intelligence without duplicating scoring or rules.

Deterministic demo path with acceptance checks

The final demo is tied to one fixed scenario and verified sample outputs.

**23**

daily-plan rows

3

recommendation rows

6

explanations

19

territory alerts

What is done, what remains

The prototype is intentionally honest about its current scope and future extensions.

HONEST LIMITS

- No production database persistence
- No auth or cloud deployment
- No live public API calls during judge demo
- NDVI and pest are reference-level foundations
- Anomaly thresholds are prototype-level

NEXT SCOPE

- Merge public signals into main feature run
- Tune anomaly thresholds
- Persist outcomes and analytics
- Add route/geospatial optimization
- Strengthen frontend testing and deployment

KshetraAI

Adaptive, explainable, measurable field execution

KshetraAI helps reps decide who to visit, what to do, why it matters, and what happened after the visit.

WHO

Who to visit

WHAT

What action to take

WHY

Why it matters

Explainable AI-assisted systems can improve agricultural field execution while keeping humans in control.